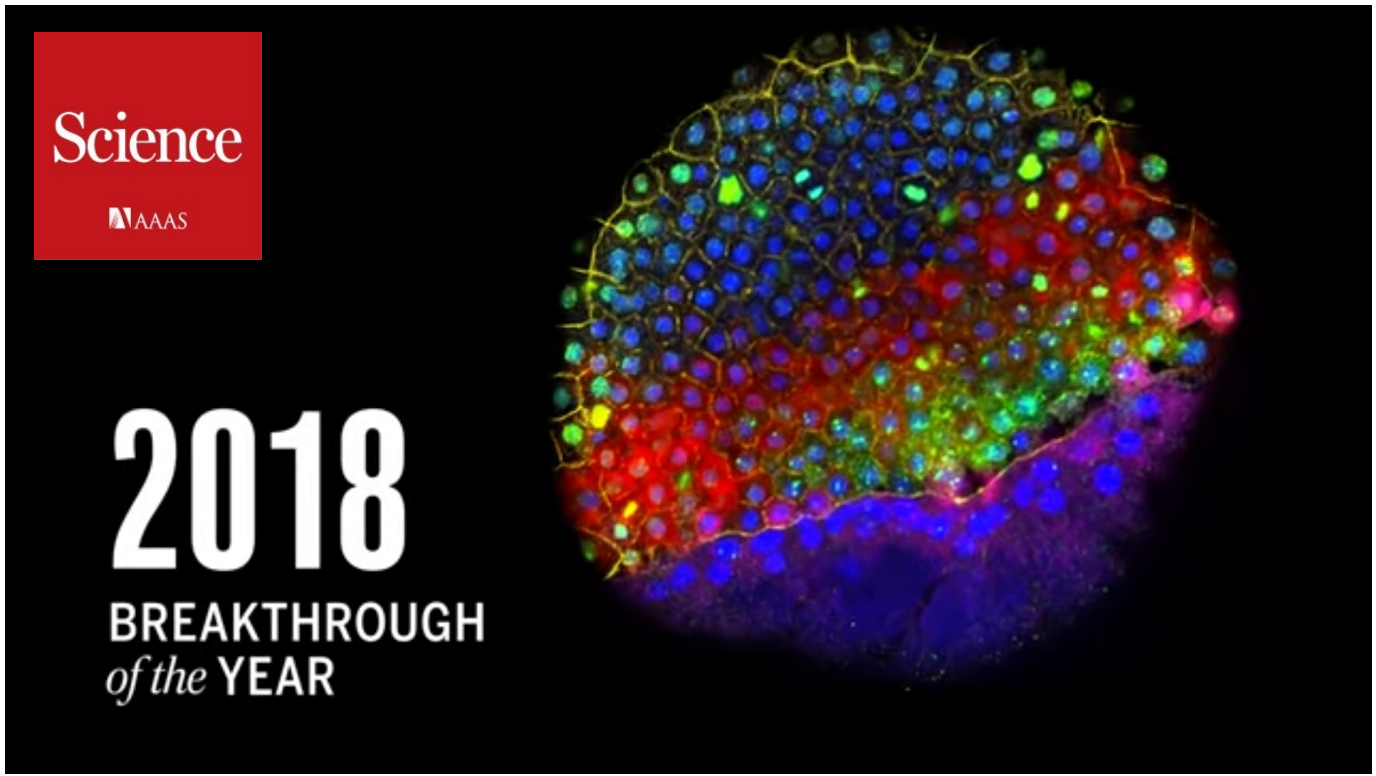
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Session 5: linear and nonlinear dimension reduction of scRNAseq data**



@NikolayOskolkov

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Personal homepage:
https://nikolay-oskolkov.com

Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Session content

Topics we'll cover in this session:

1) Dimensionality reduction: linear vs non-linear

2) Matrix factorization (PCA) vs neighbor-graph (UMAP)

3) Basic clustering: hierarchical + partitioning

4) Graph-based clustering and comparing methods

Dimensionality Reduction

1) Linear (PCA, MDS)

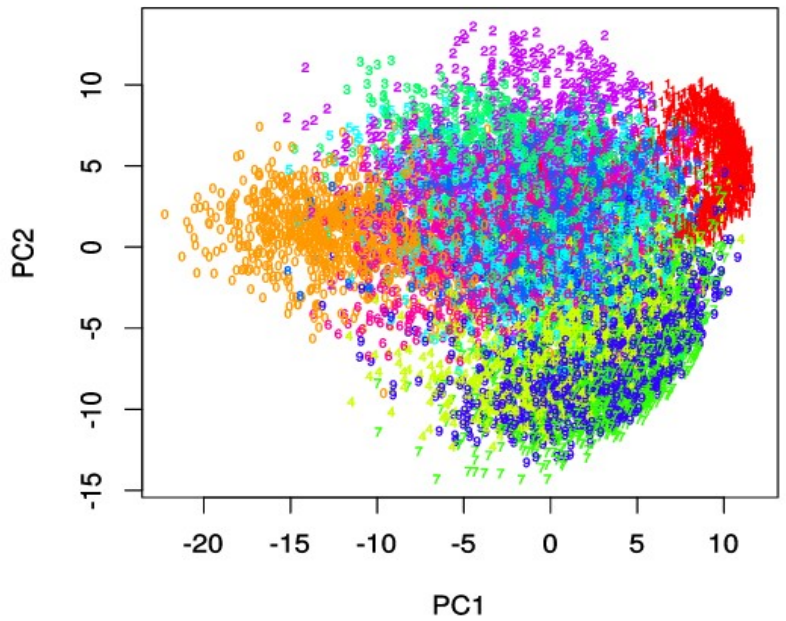
2) Non-Linear (tSNE, UMAP)

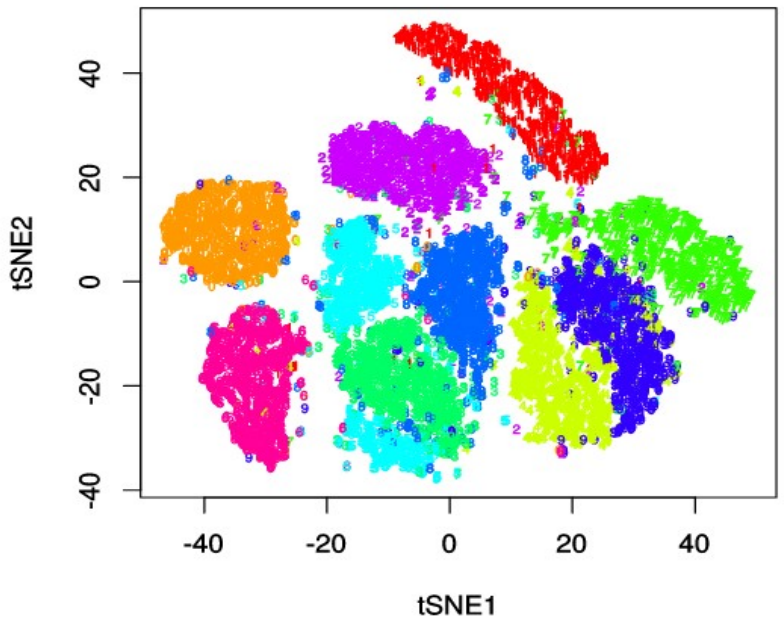
Dimension reduction: more than visualization



PCA PLOT WITH PRCOMP

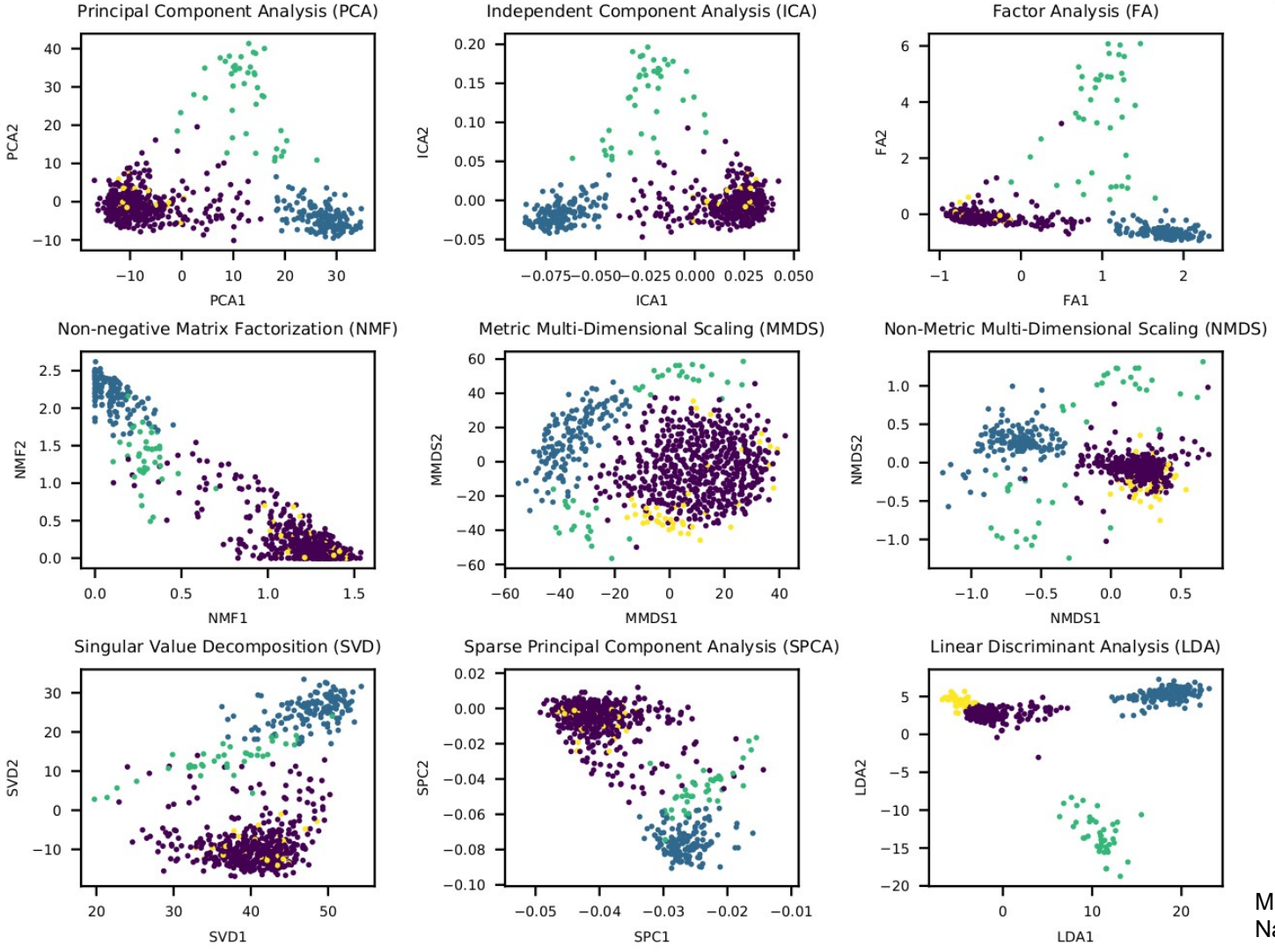
tSNE MNIST





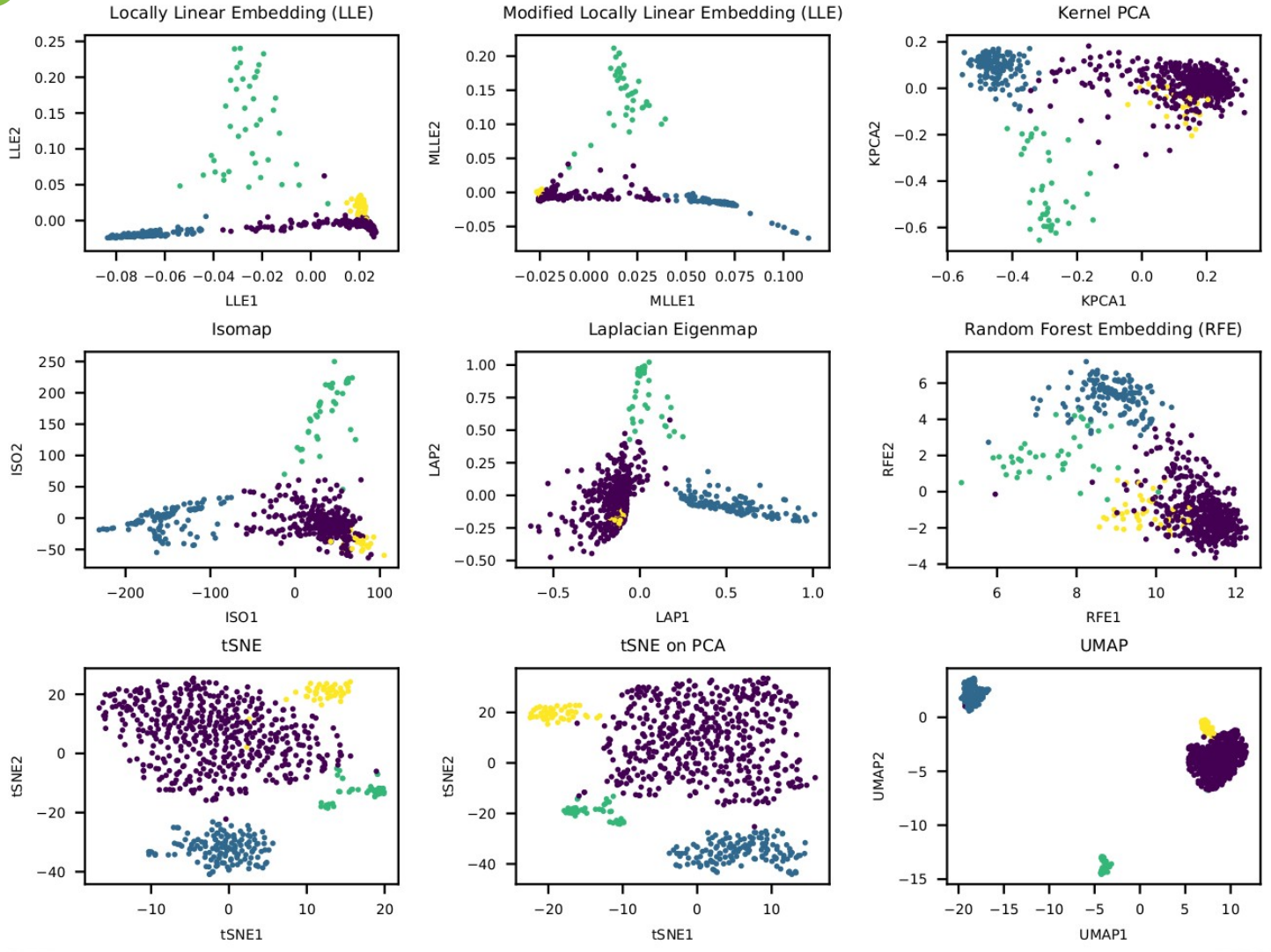
The goal of dimension reduction is not only visualization but also reducing dimensions

Linear Dimension Reduction



M.Bartoschek, N. Oskolkov et al., Nature communications 2018

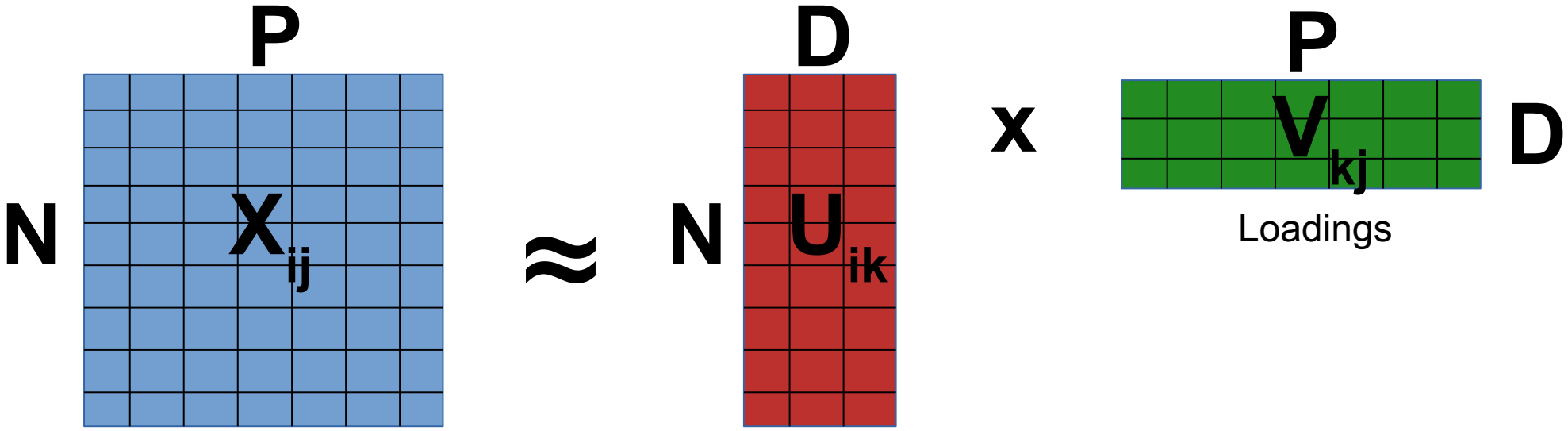
Non-Linear Dimension Reduction



M.Bartoschek, N. Oskolkov et al., Nature communications 2018

Linear dimension reduction: matrix factorization

$$X_{ij} \approx U_{ik} V_{kj}$$



Data

Low-dimensional data representation (embeddings)

Loadings

$$\text{Loss} = \sum_{i=1}^{N} \sum_{j=1}^{P} (X_{ij} - U_{ik} V_{kj})^2$$

PCA dimension reduction algorithm

Coding in R:

```
data_centered <- scale(data, center = TRUE, scale = FALSE)

covariance <- t(data_centered) %*% data_centered

eig <- eigen(covariance)

plot(eig$vectors[,1:2]);
```

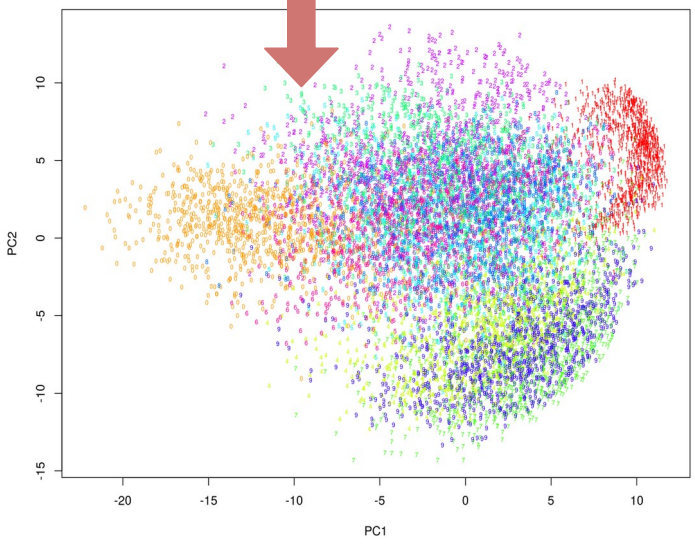
```
barplot(eig$values / sum(eig$values))
```

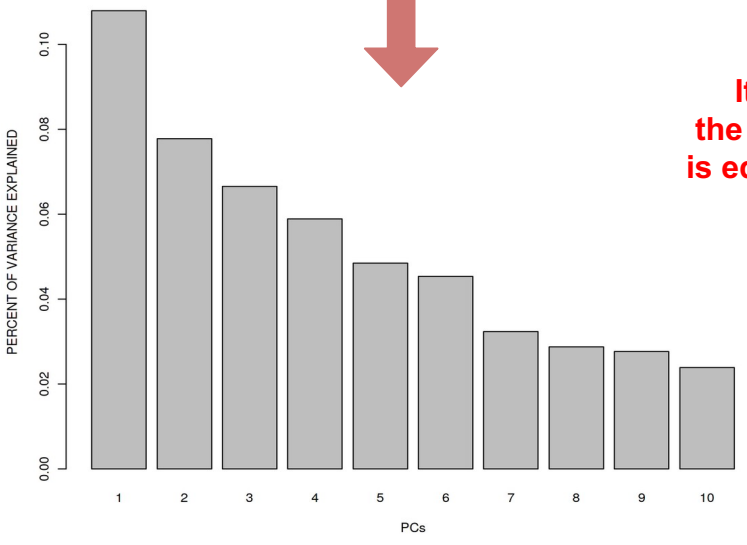
Mathematically:

$M_{ij} = X_{ij} - \mu_j$

$A = (1/N)*M^T*M$

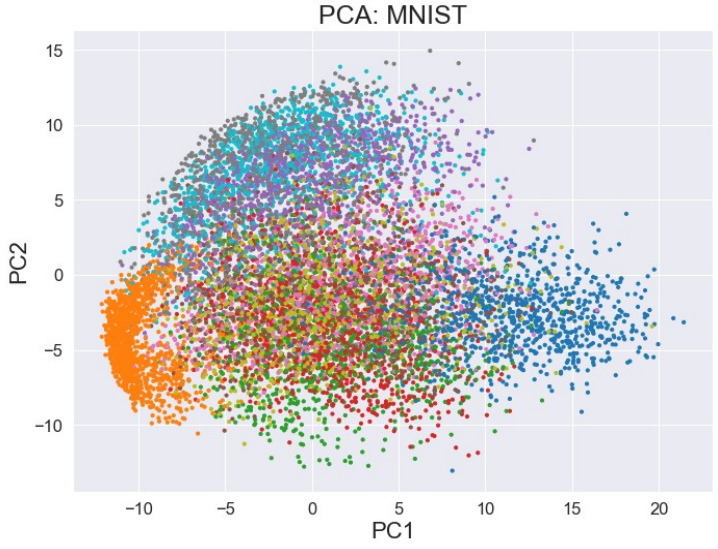
$A*u = \lambda*u$

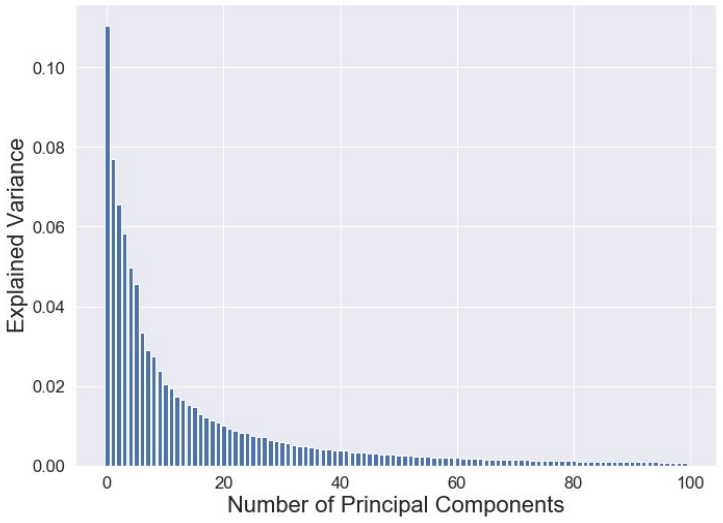


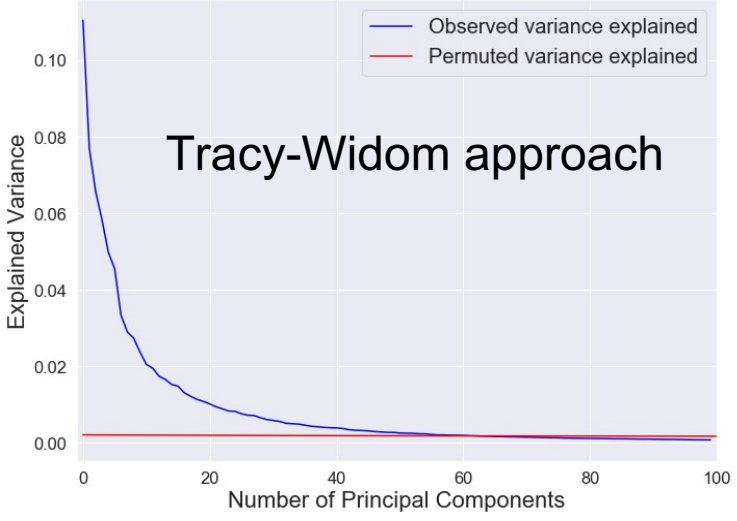


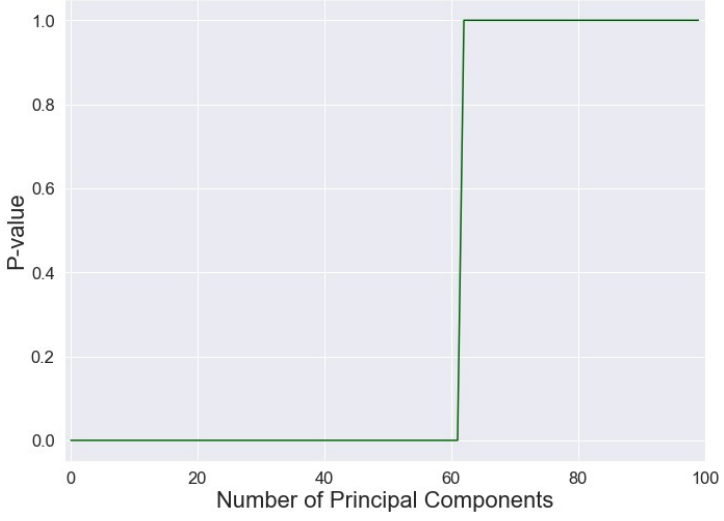
It can be analytically derived that the eigen value decomposition in PCA is equivalent to projecting data on axes of maximal variation in the data

Estimating the number of informative PCs



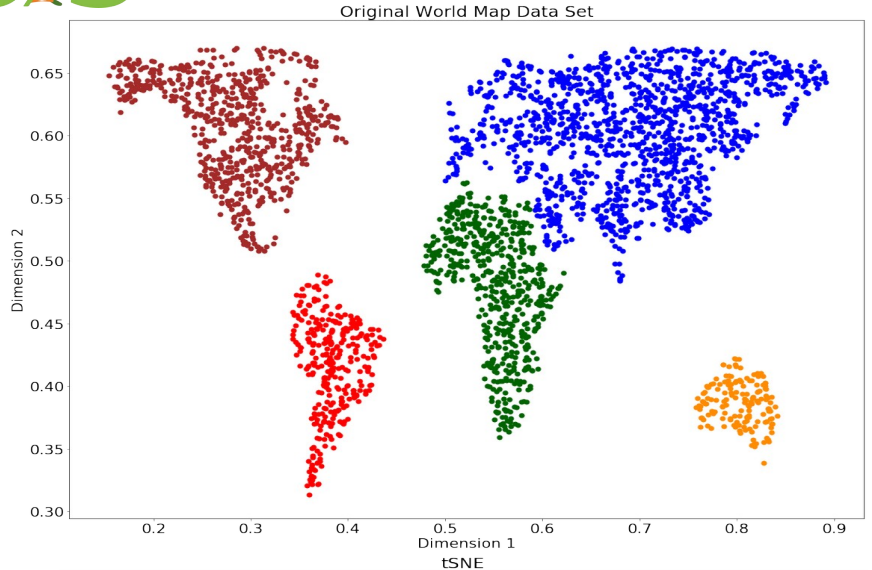


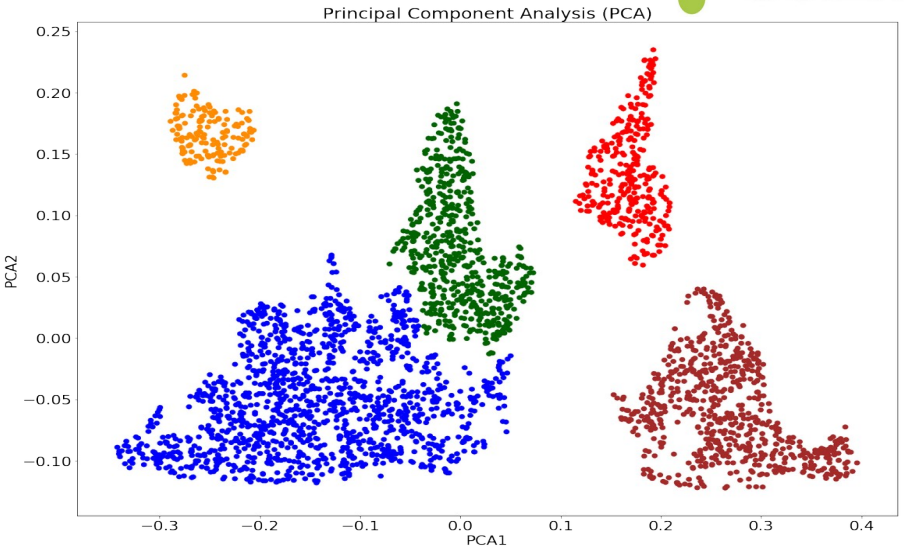


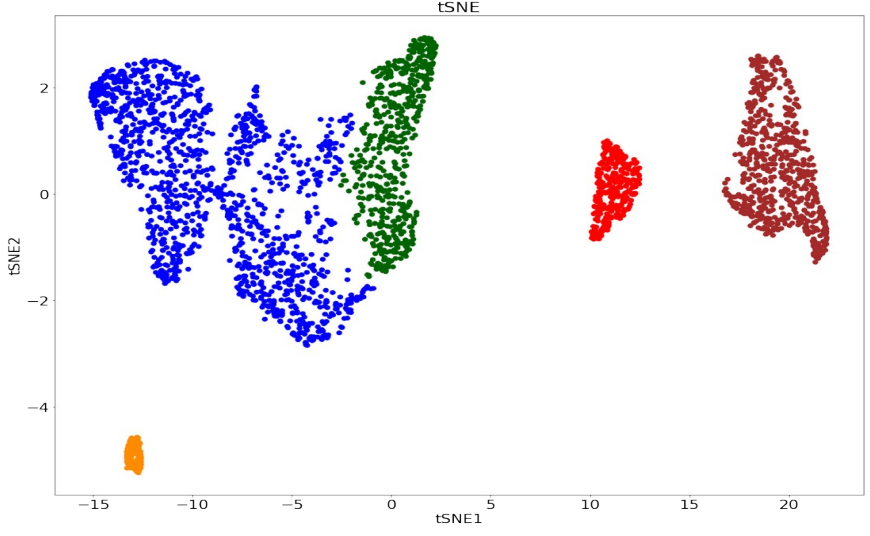


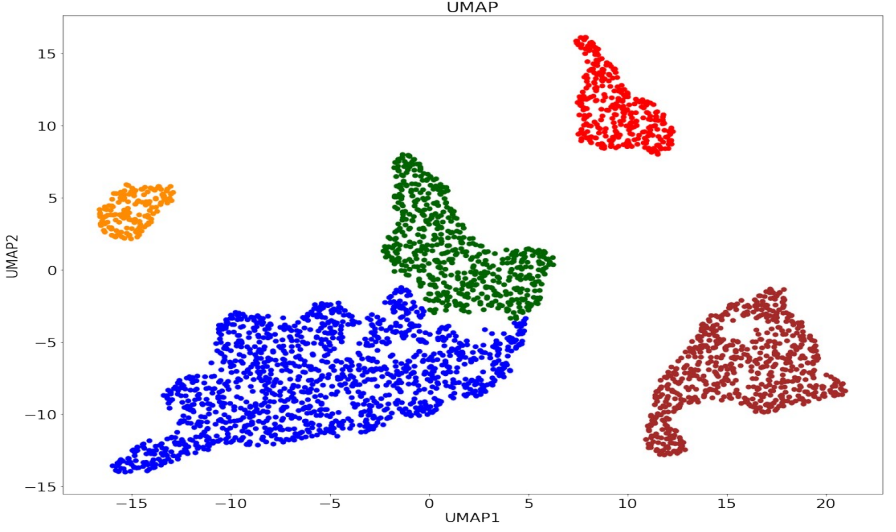
In Seurat: JackStraw

PCA works fine on a linear manifold

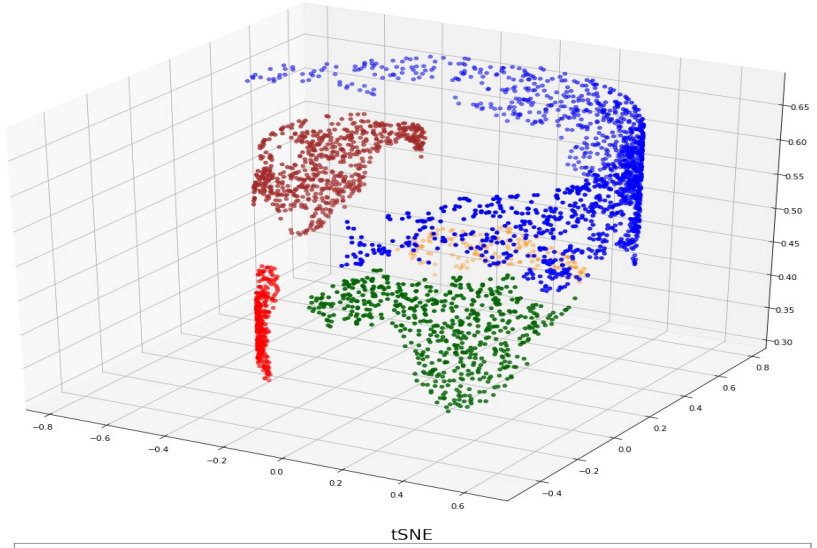


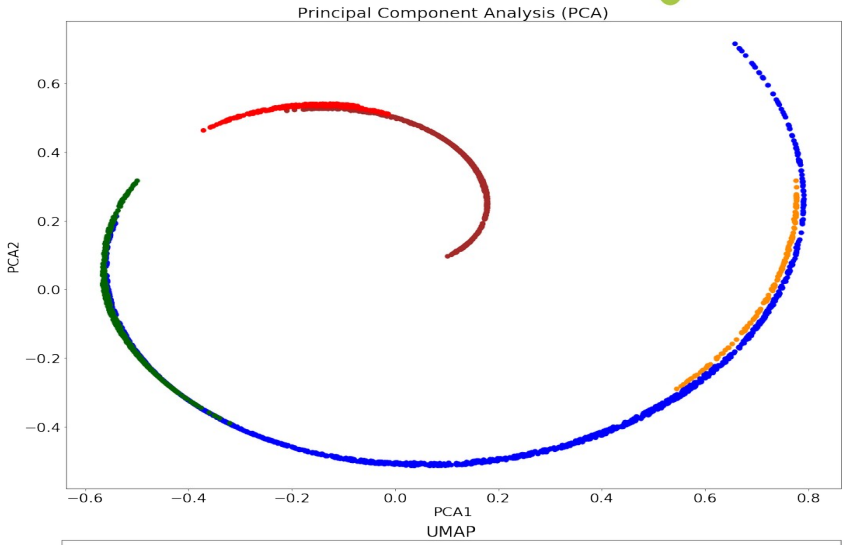


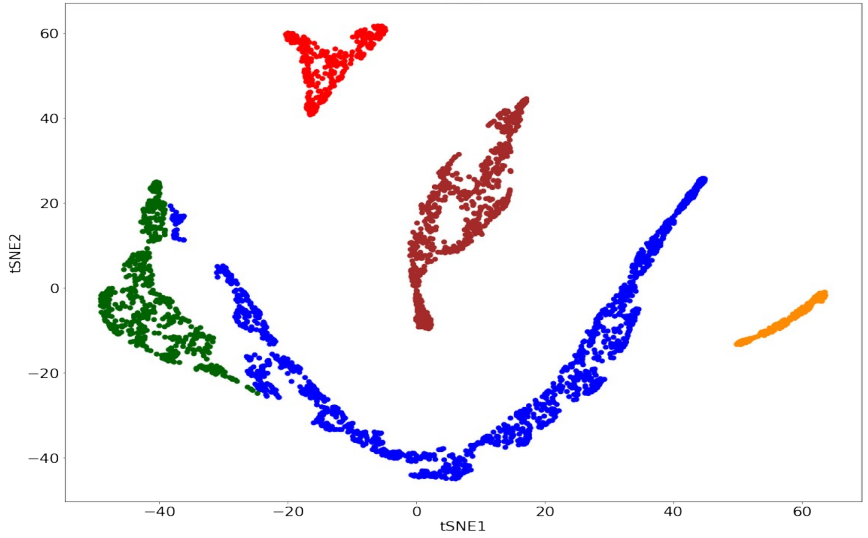


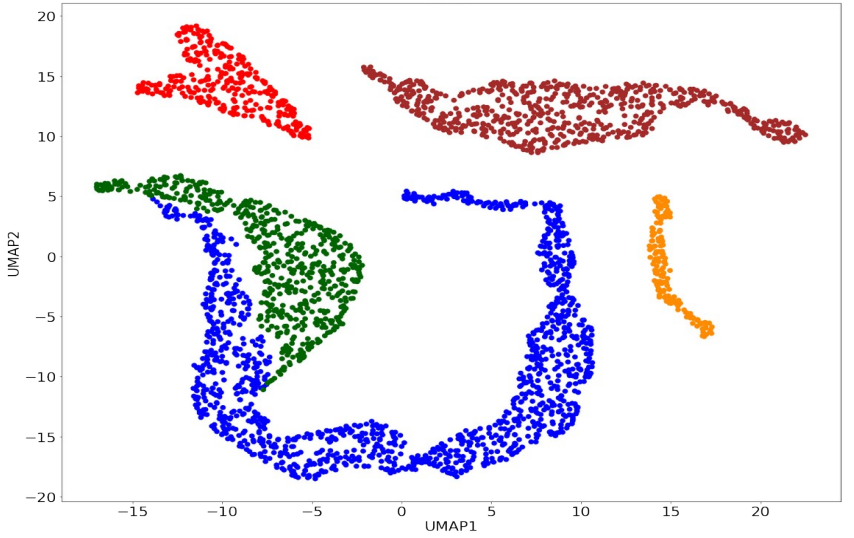


PCA vs. tSNE vs. UMAP on non-linear manifold

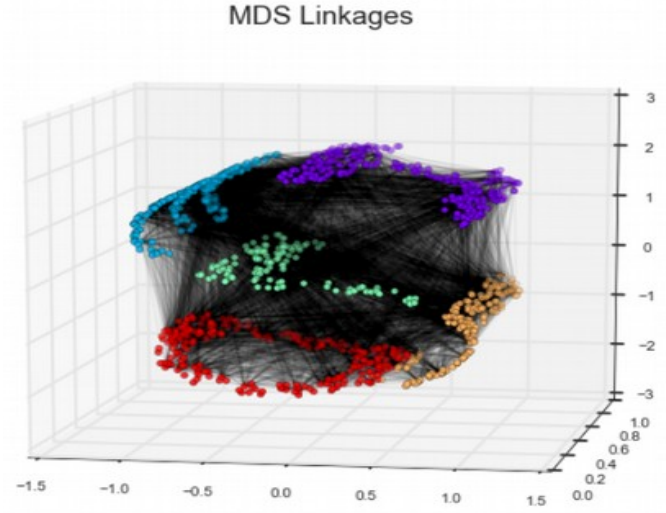


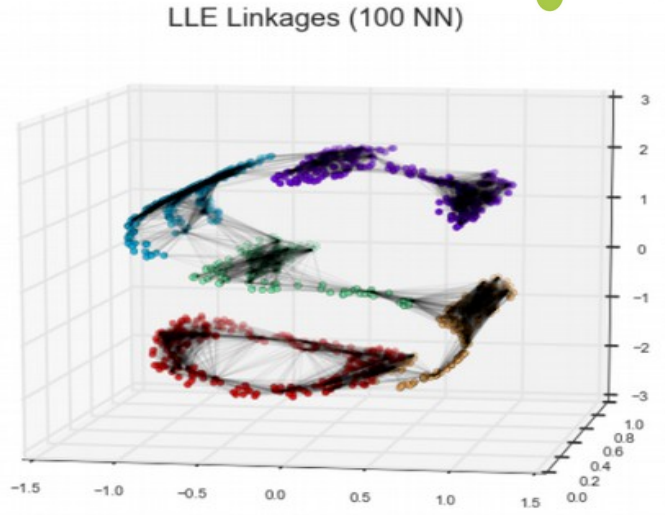


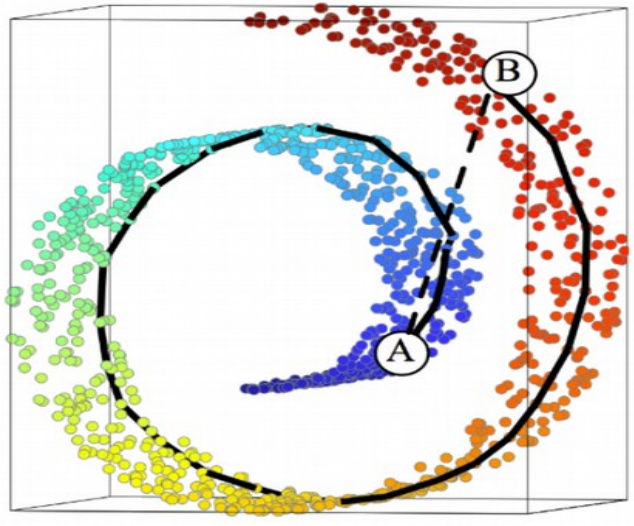


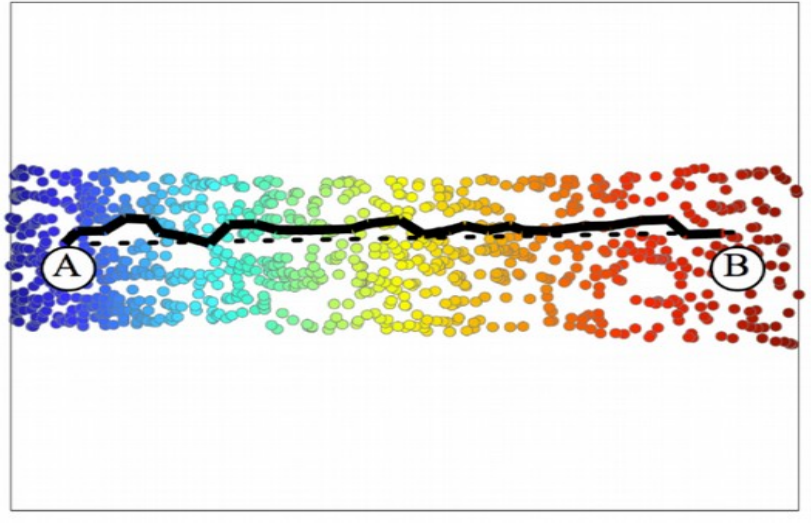


Why PCA can't unwrap the Swiss Roll



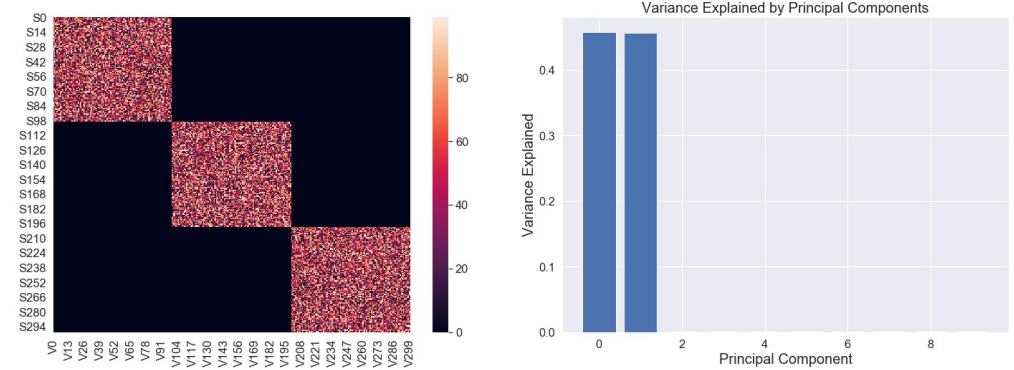


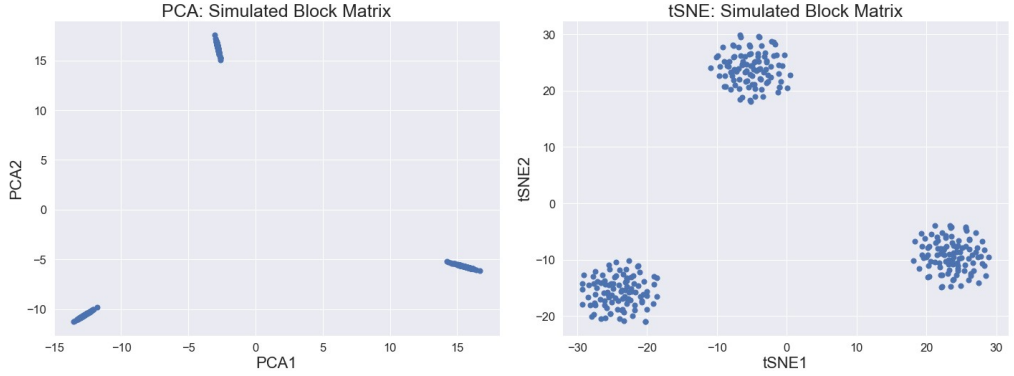




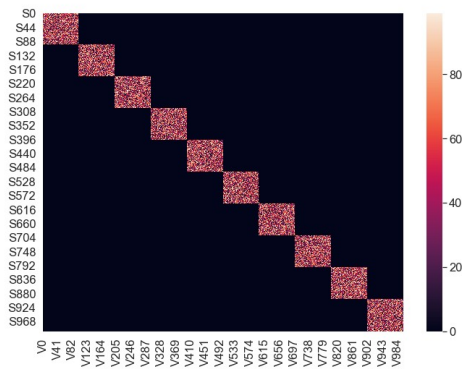
PCA vs. tSNE
when number of populations increases

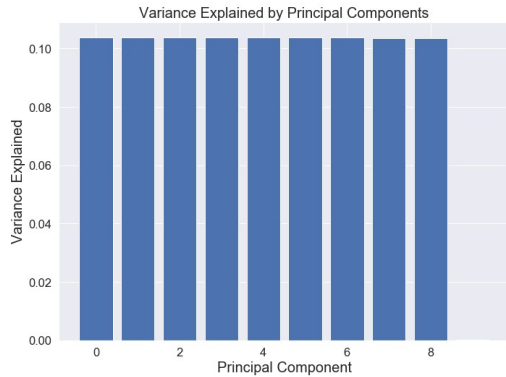
Three classes of data points

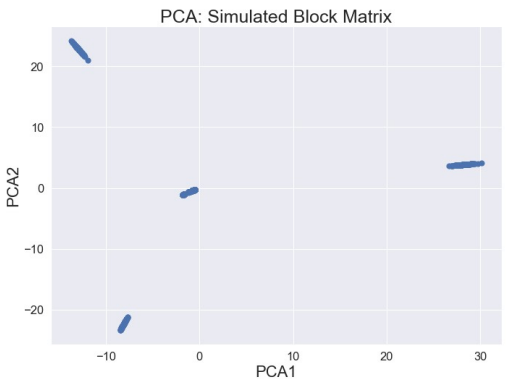


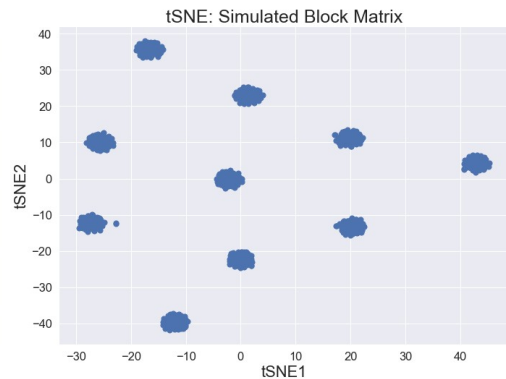


Ten classes of data points





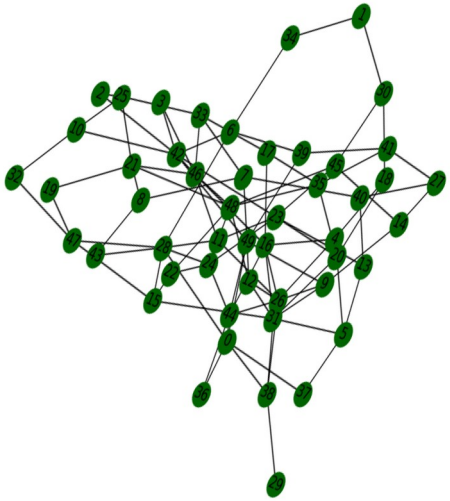




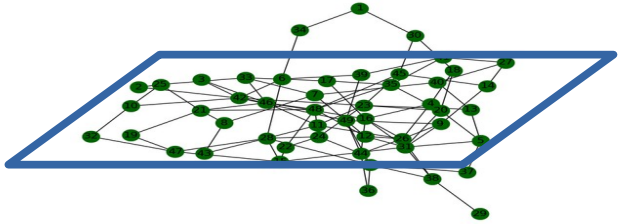
Non-linear dimension reduction: neighborhood graph

1) Construct high-dimensional graph

p_{ij}

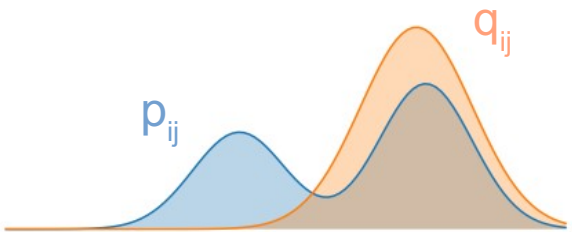


2) Construct low-dimensional graph



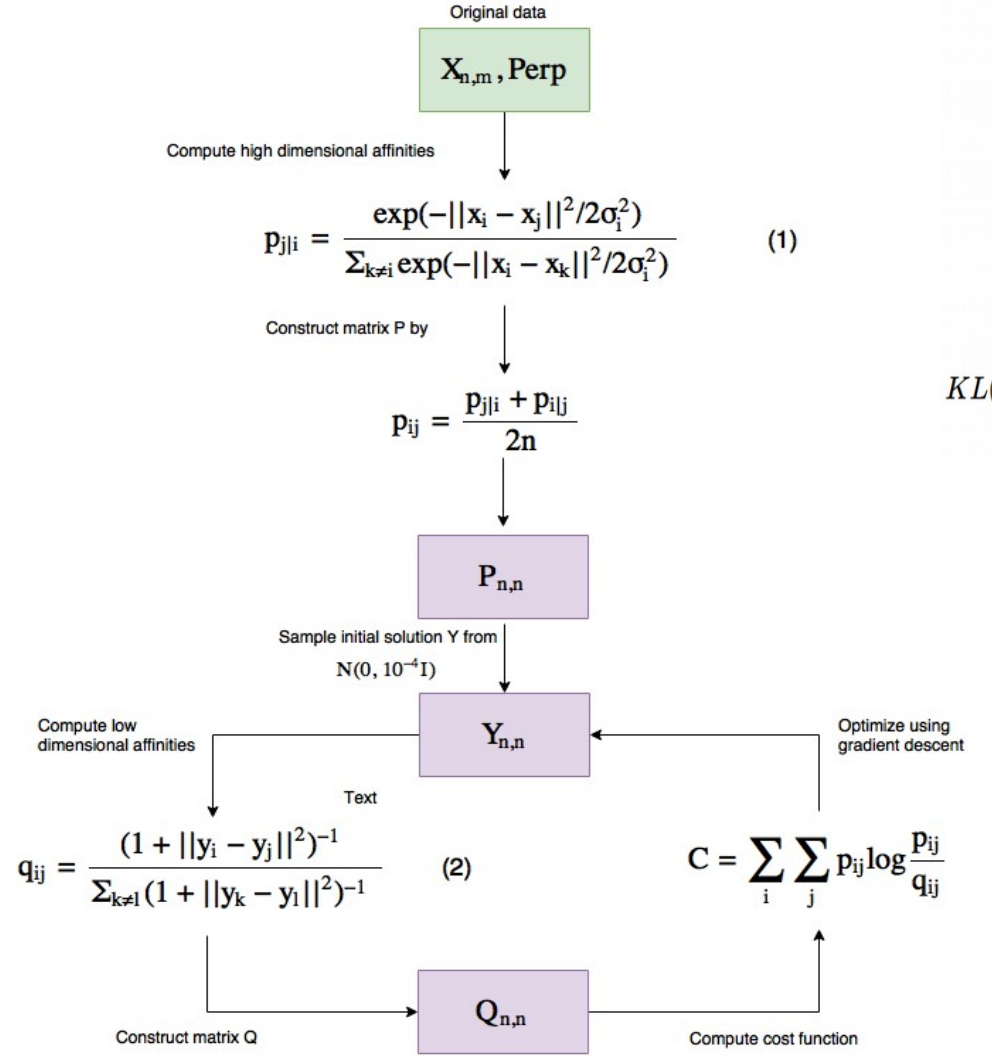
q_{ij}

3) Collapse the graphs together



Kullback-Leibler divergence

tSNE dimension reduction algorithm

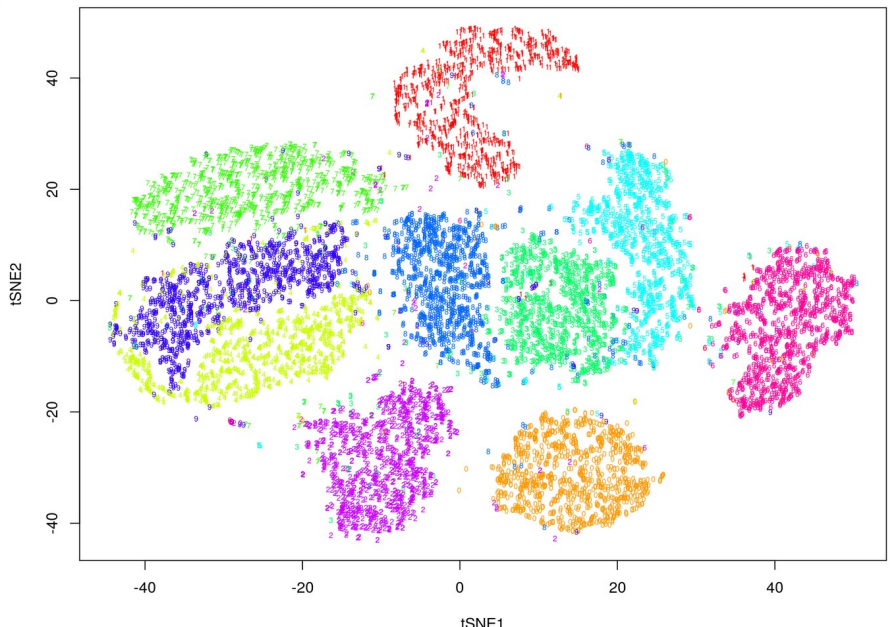


$$p_{j|i} = \frac{\exp\left(-||x_i - x_j||^2/2\sigma_i^2\right)}{\sum_{k\neq i}\exp\left(-||x_i - x_k||^2/2\sigma_i^2\right)}, \qquad p_{ij} = \frac{p_{i|j} + p_{j|i}}{2N} \tag{1}$$

$$\text{Perplexity} = 2^{-\sum_j p_{j|i}\log_2 p_{j|i}} \tag{2}$$

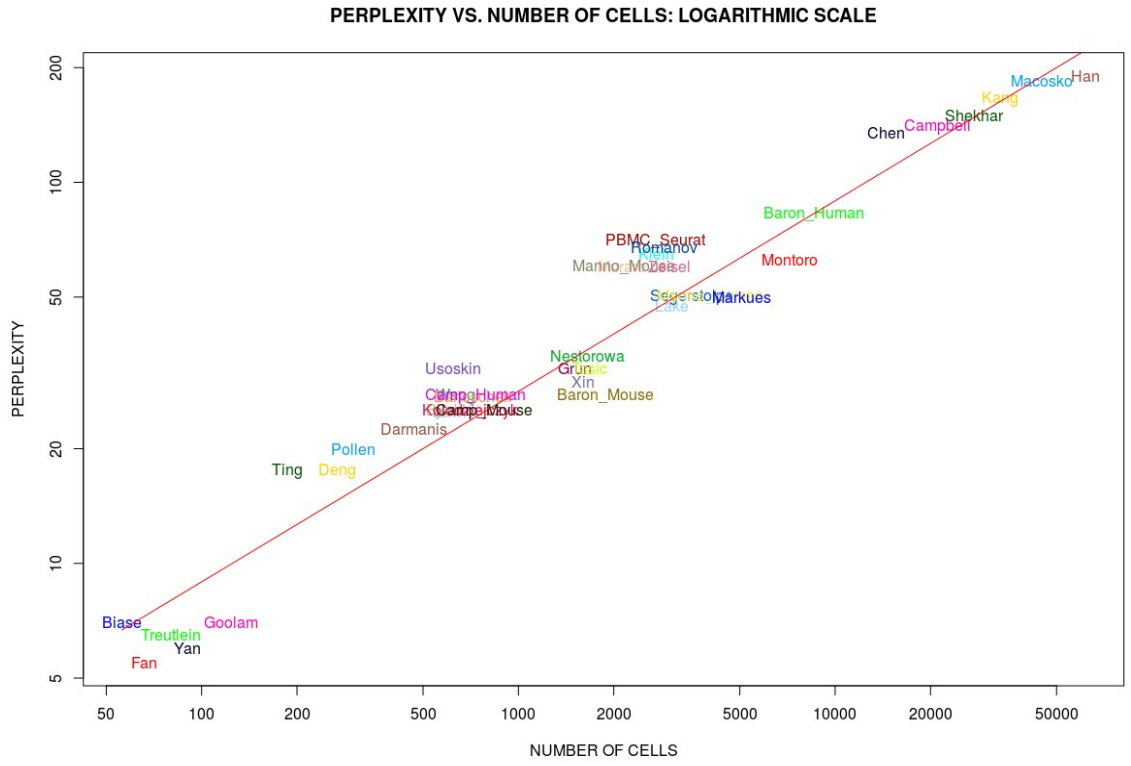
$$q_{ij} = \frac{\left(1 + ||y_i - y_j||^2\right)^{-1}}{\sum_{k\neq l}\left(1 + ||y_k - y_l||^2\right)^{-1}} \tag{3}$$

$$KL(P_i||Q_i) = \sum_i \sum_j p_{j|i}\log\frac{p_{j|i}}{q_{j|i}}, \qquad \frac{\partial KL}{\partial y_i} = 4\sum_j (p_{ij} - q_{ij})(y_i - y_j)\left(1 + ||y_i - y_j||^2\right)^{-1} \tag{4}$$



How to select optimal perplexity

Van der Maaten: “Loosely speaking, one could say that a larger / denser dataset requires a larger perplexity.“



$$\log(Perp) = -0.179 + 0.51*\log(N)$$

$$Perp \sim N^{(1/2)}$$

Limitations of tSNE and promise of UMAP

tSNE does not scale for large data sets?

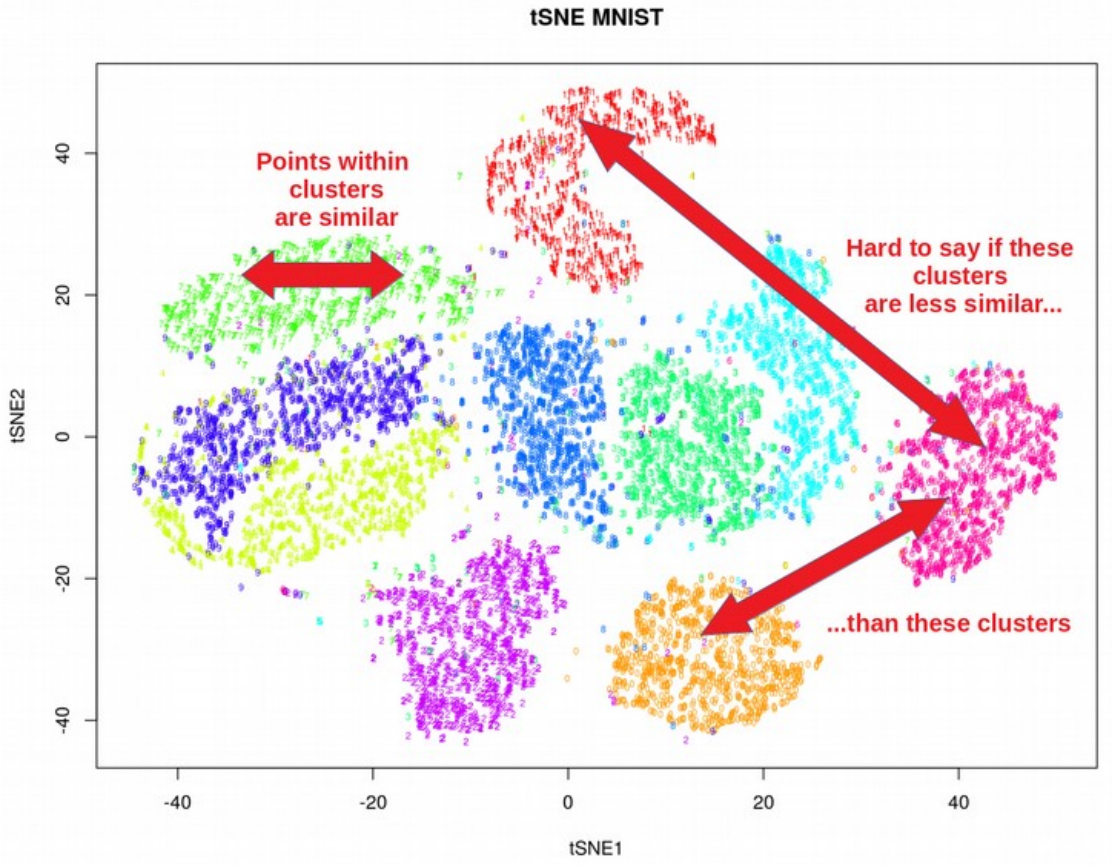
tSNE does not preserve global structure?

tSNE can only embed into 2-3 dims?

tSNE performs non-parametric mapping (no variance explained statistics)?

tSNE can not work with high-dimensional data directly (PCA needed)?

tSNE uses too much RAM at large perp?



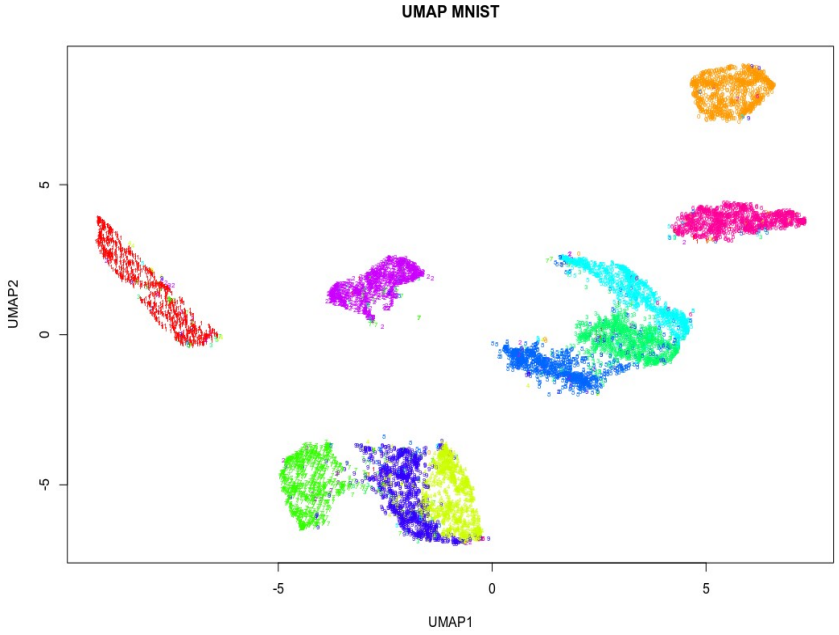
How is UMAP different from tSNE

UMAP uses local connectivity for high-dim probabilities

$$p_{i|j} = e^{-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}}$$

UMAP does not normalize probabilities (speed-up)

UMAP can deliver a number of components for clustering



UMAP uses Laplacian Eigenmap for initialization

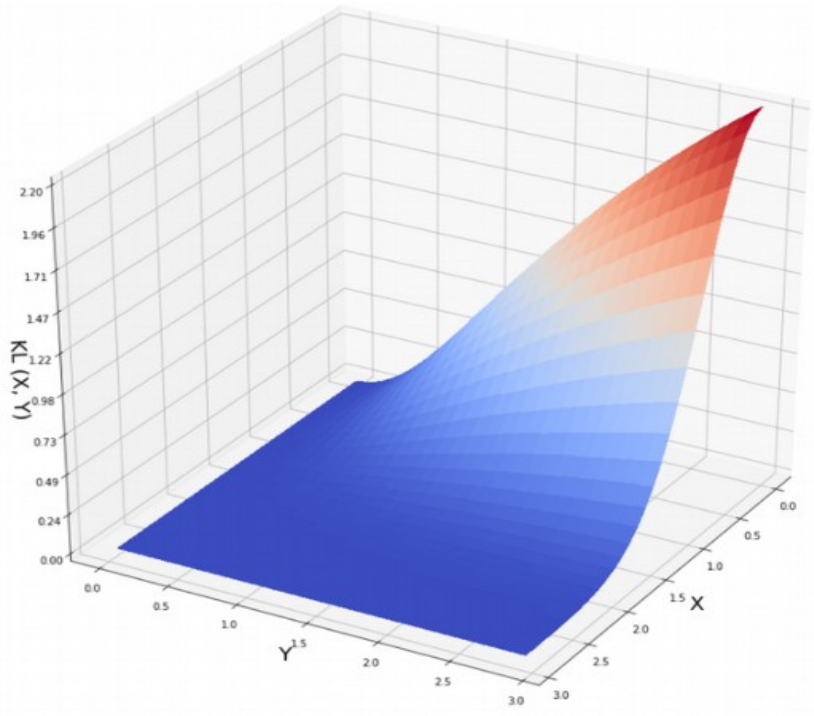
UMAP uses Cross-Entropy (not KL) as cost function

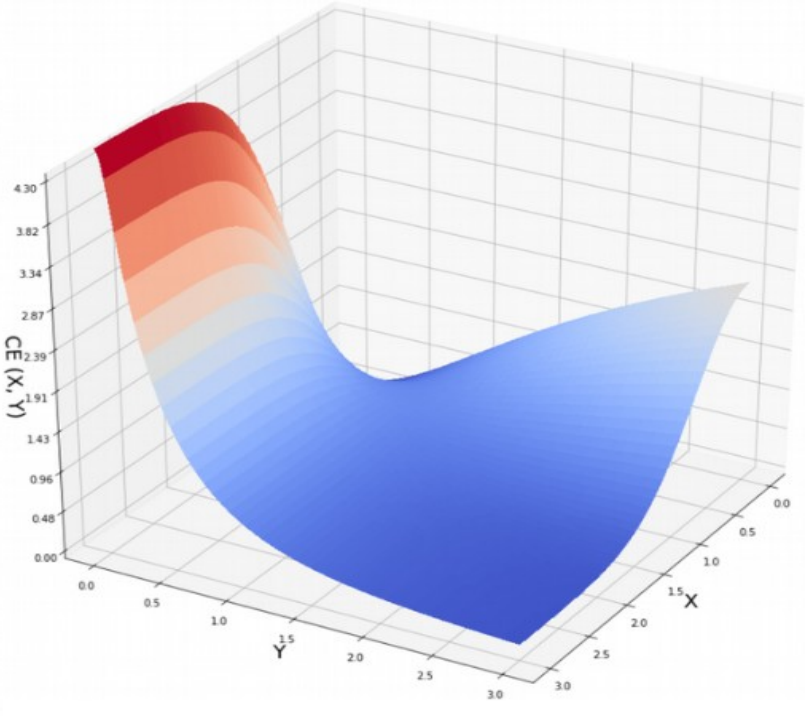
$$CE(X,Y) = \sum_i \sum_j \left[p_{ij}(X) \log\left(\frac{p_{ij}(X)}{q_{ij}(Y)}\right) + (1 - p_{ij}(X)) \log\left(\frac{1 - p_{ij}(X)}{1 - q_{ij}(Y)}\right) \right]$$

This is similar to tSNE cost function

This term is UMAP specific

Cost Function Makes UMAP Preserve More Global Structure than tSNE

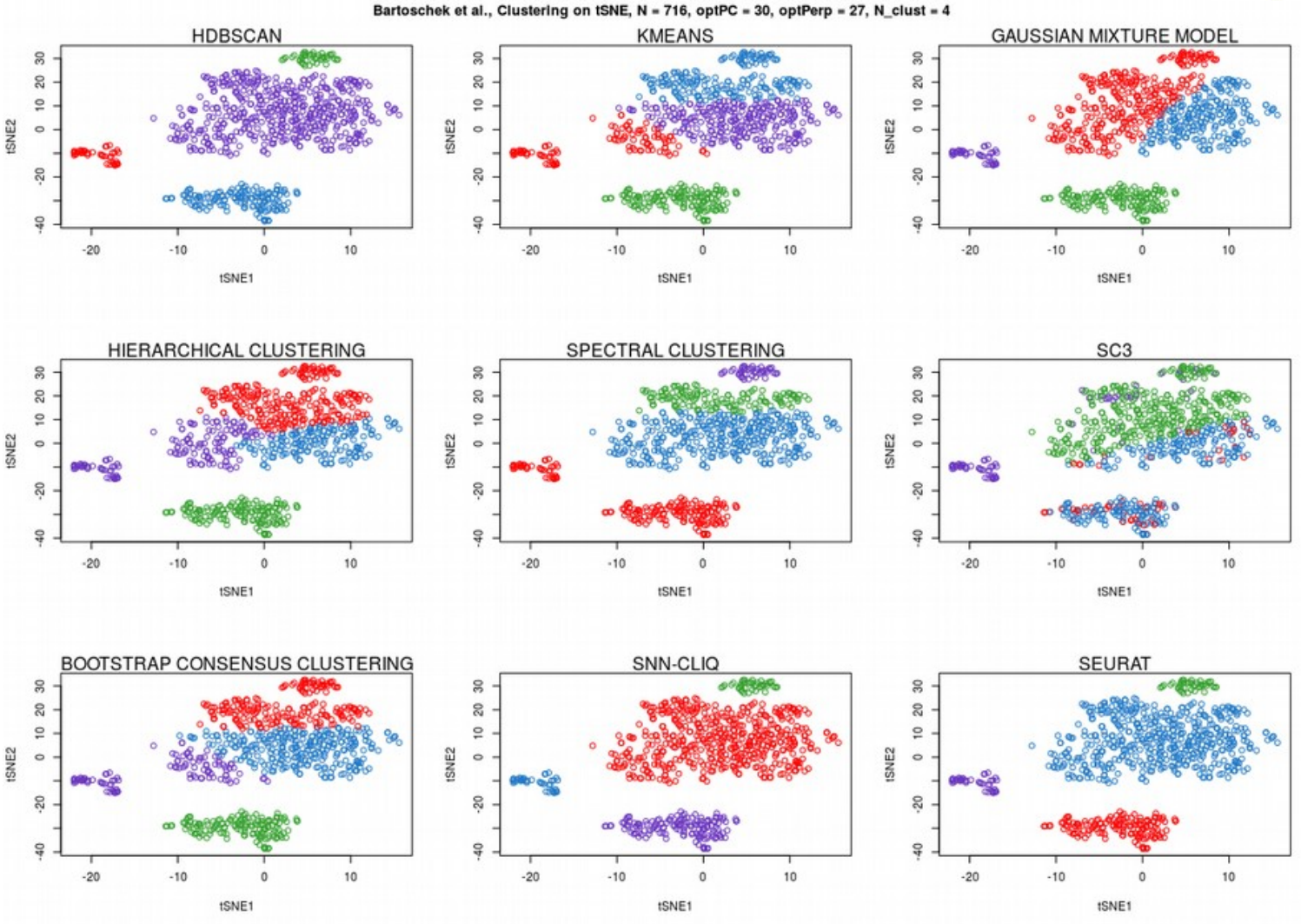




X → infinity, Y can be any

X → infinity, Y → infinity

Clustering on two tSNE dimensions



Session summary

Take home messages of the session:

1) Linear dimensionality reduction (PCA, MDS) does not have enough resolution for scRNAseq data

2) Non-linear dimension reduction (tSNE and UMAP) is advantageous for complex single cell data

3) UMAP preserves more of a global structure in single cell RNAseq data

National Bioinformatics Infrastructure Sweden (NBIS)



Knut och Alice Wallenbergs Stiftelse

Vetenskapsrådet

Lunds Universitet

Acknowledgments: NIRI + TARGETWISE



European Regional Development Fund project “Functional Omics Analysis of Metabolic Diseases to Advance Drug Discovery Research Excellence in LIOS - TARGETWISE” (No.1.1.1.5./2/24/A/003)

2027 National Development Plan

Funded by the European Union

Acknowledgments: Metabolic Research Group



Open positions: fully covered PhD and 3-year postdoctoral fellow positions!